

# Comparison between Multinomial Naive Bayes and Multi-Layer Perceptron for Product Review In Real Time

Tanmay Singla<sup>1</sup>, Vaibhav Gaur<sup>2</sup>, Devendra Kumar Misra<sup>3</sup>

<sup>1,2,3</sup> Information Technology Department, HMR Institute of Technology and Management, Hamidpur, New Delhi, India

<sup>1</sup>tanmay1903@hotmail.com

<sup>2</sup>vaibhavgaur45@gmail.com

<sup>3</sup>misra.deven@gmail.com

**Abstract**— To find out and understand users' requirements, complaints, and preferences, user feedback can be used as a perfect tool. Taking into account the amount of user feedback received on a product every day, it is difficult for the developers to identify issues that users have for the product or services from the user-given feedback, as it takes a lot of time and thus becomes very inefficient. To overcome this problem, an automation system is used to fetch the real-time product reviews and then a comparison between Multinomial Naïve Bayes and Multi-Layer Perceptron was built to determine which algorithm performs better with a dataset of a real time product review. Document/Text Classification and language representation in vector form is useful in the field of machine learning. In this paper, we aim to use the Text Classification and vector representations method to decode the reviews gathered in real-time into supervised form and then apply the Multinomial Naïve Bayes and Multi-Layer Perceptron classification algorithm to evaluate the Sentiment Analysis on the website products.

**Keywords**— Text Classification, Multinomial Naïve Bayes, Multilayer Perceptron, vector representations, sentiment analysis, user feedback, Real-Time

## I. INTRODUCTION

E-Commerce websites contain millions of products, which users can order as per their needs. Users can give their views and sentiment toward Products, which will then have an influence on other users' perspectives and decisions and help both service providers and developers to evaluate their quality.

Firstly, we collected real-time product reviews from Amazon. and Flipkart.com using Automation, and then the data was pre-processed before vectorization. In addition, we created a sparse matrix to get a better understanding of the frequency of occurrence. Secondly, we compared the accuracy and performance of Multinomial Naïve Bayes and Multi-Layer Perceptron using the Confusion Matrix. Lastly, we compared the accuracy and showed the change in the ROC curve [1].

Recognizing issues from the client's inputs because of their attributes can be challenging. First and foremost, client criticism contains various clamor words, like incorrectly spelled words, redundant words, also, shoptalk words, which is a famous nearby language among Indian adolescents. These things make it trying to arrange whether the client is fulfilled

or disappointed with the item. Also, for a few famous items, the volume of client inputs produced every day is huge to be physically perused and broken down. In conclusion, not all criticism contains data that is valuable. Just 33% of client criticisms contain data that can straightforwardly assist with adjusting suppliers and engineers to further develop their service.

To dissect the feeling of given client criticism, the submitted item survey should be pre-handled utilizing regular language handling methods. For text written in English, there are a few NLP tool compartments accessible, for example, Stanford Core NLP Toolkit, OpenNLP, or NLTK. Nonetheless, a coordinated NLP tool compartment for Indian Language is as yet uncommon; or possibly, this is the situation at the time our examination was led. In this way, we chose to web creep the survey from other online business sites to advance our information with Indian Local Languages [2].

As an initial move towards an exhaustive help instrument for examining client inputs, this exploration proposes an apparatus to order client criticisms as per its general feeling naturally. AI classifiers have been generally utilized with the end goal of opinion investigation giving great outcomes. Among different AI calculations, the Multinomial Naïve Bayes (NB) calculation is for the most part utilized for the characterization issue because of its effortlessness and adequacy.

Multinomial Naïve Bayes classifier works on the concept of term frequency which means how many times the word occurred in a review. This model provides two factors whether the expression occurred in a reviewer or not, as well as its number of occurrences in that review. Anticipating the polarity as 0 or 1 of a product review, the probabilities are multiplied by the frequency of all the expressions in the product review against one and the other polarities and the one with the larger value gives the analysis of this product review.

## II. LITERATURE REVIEW

Sentiment Analysis means to understand users' requirements, complaints, and preferences from users' feedback and analyze on the basis of positive, negative and neutral sentiment. Previous papers used other probabilistic

learning methods other than Multinomial Naive Bayes and Multi-Layer Perceptron, like Bernoulli Naïve Bayes, Support Vector machines, word vectors etc. to find sentiments of offline or pre-downloaded reviews data of either movies, services and products etc. The studies show that Multinomial Naive Bayes and Multi-Layer Perceptron, both work great for sentiment analysis and text classification. But the limitation in these studies is that the algorithms were trained and tested on static data of reviews, not the real-time. With Time, a lot of things change and so do users' sentiment towards a product or service. So, it becomes necessary to research about these algorithms' accuracy and efficiency with the real-time data of reviews.

The paper is organized as follows: Section III explains the methodology, the dataset and the classification models used to go about this research to analyze reviews. Section IV shows the model evaluation and results i.e. the accuracy and efficiency of both the algorithms on real-time data. Finally, Section V concludes the work that has been proposed and its future scope.

### III. METHODOLOGY

#### A. Automation

Automation is done using Selenium WebDriver, which is used for automating web applications for testing purposes. Automation is done on different E-Commerce websites to retrieve user reviews for training the algorithm and thereafter retrieving the Realtime User feedback of a particular product to test and analyze the sentiments of the users regarding the product on various E-Commerce Websites.

```
from pyvirtualdisplay import Display
from selenium import webdriver
from fake_useragent import UserAgent
display = Display(visible=0, size=(800, 600))
display.start()

chrome_options = webdriver.ChromeOptions()
ua = UserAgent()
userAgent = ua.random
chrome_options.add_argument('--disable-extensions')
chrome_options.add_argument(f'user-agent={userAgent}')
chrome_options.add_argument('--profile-directory=Default')
chrome_options.add_argument("--incognito")
chrome_options.add_argument("--disable-plugins-discovery")
chrome_options.add_argument("--start-maximized")
driver = webdriver.Chrome()
```

Fig. 1 Snapshot of Selenium WebDriver For Automation

Figure 1 is a snapshot of the python code used to integrate Selenium WebDriver which is then used for Automating the other websites like Amazon, and Flipkart for user reviews for any product listed on those websites.

#### B. Dataset

The dataset comprises 1596 reviews of Amazon which has twenty-seven columns out of which we only needed two columns namely the title which has the brief of the review and

'rating' which let us know about the tendency of the review in terms of rating stars out of five[3].

|   | reviews.title                              | reviews.rating |
|---|--------------------------------------------|----------------|
| 0 | Paperwhite voyage, no regrets!             | 5.0            |
| 1 | One Simply Could Not Ask For More          | 5.0            |
| 2 | Great for those that just want an e-reader | 4.0            |
| 3 | Love / Hate relationship                   | 5.0            |
| 4 | I LOVE IT                                  | 5.0            |

Fig. 2 Snapshot of Data Under Analysis

Figure 2 is a snapshot of the dataset of user reviews that are used to train the analysis model.

#### C. Data Preprocessing

After Automating the data, it's been loaded into the workspace for further processing, the text needs to be cleared and pre-processed to achieve finer precision [3]. The Data is pre-processed to prepare the data so that it is more successful in producing the intended result while building a model by removing the unwanted attributes in the data as shown in Figure 3[12-18].

```
['paperwhit voyag regret', 'one simpli could ask', 'great want e r  
ller size', 'superb read devic one best', 'love', 'un plaisir', 'wc  
d easi find need', 'seamless reas experi', 'great addit kindl rang'  
und help review help ye', 'kindl vs nook updat peopl found help rev  
sh novel good light peopl found help review help ye', 'worth money  
ound help review help ye', 'want dedic e reader got peopl found hel  
d help review help ye', 'work great one critic flaw significantli i  
changer demand content amazon knock park refresh updat tip hbo go f  
ritic flaw significantli improv peopl found help review help ye', '  
'box game changer demand content amazon knock park refresh updat ti  
mazon import peopl found help review help ye amazon com offici comm  
view help ye', 'box game changer demand content amazon knock park r  
lp ye', 'like amazon import peopl found help review help ye amazon  
opl found help review help ye amazon com offici comment review', 'l  
efresh updat tip hbo go peopl found help review help ye', 'great up  
sign flaw', 'major fan buy protect', 'fabric cloth', 'love', 'great  
'perfect lot peopl comment', 'great upgrad aesthet', 'updat review  
o echo echo love tap', 'kf hdx ok homework prime download buy', 'er  
till batteri hog though', 'dead month', 'everyth expect', 'kf hdx c
```

Fig. 3 Snapshot of the preprocessed data

The following steps are involved in the Basic Data pre-processing:

##### a). Tokenization:

It is a process in which an entire text is converted into sets of tokens with each token being unique.

##### b). Lowercasing:

Each alphabet is converted into lowercase to attain uniformity so that even if the same word was written in Lowercase or Uppercase it won't be handled individually.

##### c). Stopwords Removal:

Several expressions in the tokenized text do not deem any extensive statistics but may impose a notable impression on the classification model. It is advised to take out those expressions in advance if the dataset is greater.



of the loss function w.r.t a parameter that requires alteration, i.e.

$$ww \leftarrow \eta \left( \alpha \frac{\partial R(w)}{\partial w} + \frac{\partial Loss}{\partial w} \right) \quad (4)$$

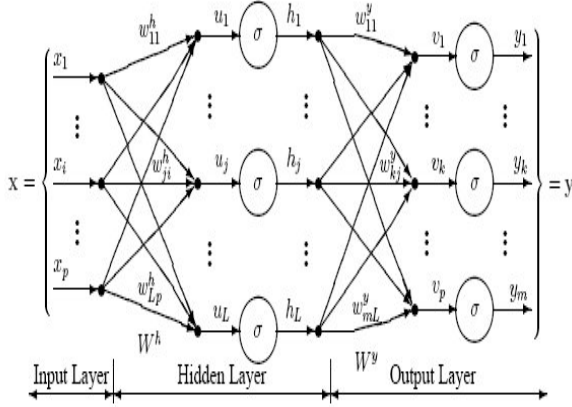


Fig. 6 Multi-Layer Perceptron Algorithm Flow

In Equation 4,  $\eta$  :regulates the step size in the parameter space search and represents the learning rate. Loss : represents the loss function.

Given a set of training examples of product review as  $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$  where  $x_i \in \mathbb{R}^n$  and  $y_i \in \{0, 1\}$ , a one hidden layer and one hidden neuron Multi-Layer perceptron learns the function  $f(x) = W_2 g(W_1 T x + b_1) + b_2$  where  $W_1 \in \mathbb{R}^m$  and  $W_2, b_1, b_2 \in \mathbb{R}$  are model parameters. The weights of the input and hidden layer are represented as  $W_1, W_2$ , respectively[10].

The activation function is represented as  $g(.) : \mathbb{R} \rightarrow \mathbb{R}$ , formulated as,

$$g(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}} \quad (5)$$

In Eqn. 5,  $f(x)$  propagates through the  $g(z) = 1/(1+e^{-z})$  which is a logistic function, for binary classification, to derive output values in the range of 0 and 1. A threshold value is set at 0.5, a positive class is assigned to the output values which are greater than or equal to the threshold, and the rest to the negative class.

MLP uses Cross-Entropy as the loss function for the classification of the binary case, Equation 6 which helps in predicting the sentiment of the review as positive or negative. Cross-Entropy can be formulated as :

$$Loss(\hat{y}, y, W) = -y \ln \hat{y} - (1 - \hat{y}) + \|W\|_2^2 \quad (6)$$

Initially, random weights are taken, and MLP reduces the loss function by constantly improving these weights. After calculating the loss at the output layer, backward propagation is done to improve the weights in preceding layers of neurons meant to decrease the loss.

Gradient descent is formally expressed as,

$$M^{j+1} = M^j - \epsilon \nabla Loss_M^j \quad (7)$$

M: represents the weights

$\nabla Loss_M$ : gradient of the loss with respect to the weights(M)

j : represents the next repetitive steps

$\epsilon$  : represents the learning rate with a natural number.

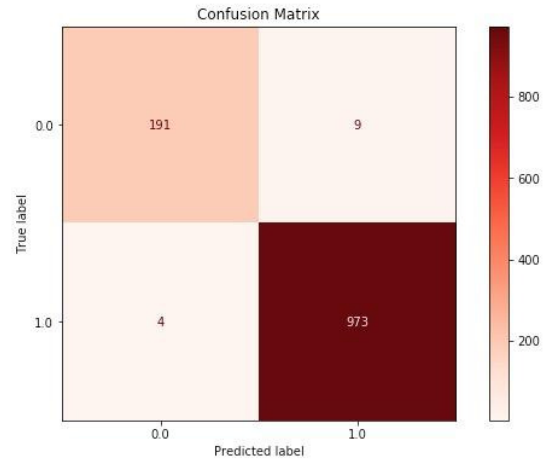
The algorithm halts as it approaches a preset number of repetitive steps; further refinement in loss can't be done below a certain small number.

#### IV. MODEL EVALUATION & RESULTS

Results obtained from the Multi-Layer Perceptron and Multinomial NB(Naïve Bayes) on the dataset of product reviews to date automated from flipkart.com and amazon. are summarized in this section. Observation shows that on the given dataset Multi-Layer Perceptron performs more excellently and effectively than Multinomial Naïve Bayes. Essential models were trained by 1596 features.

Confusion matrix for Multi-layer Perceptron (MLP) is shown in Fig.-7a which represents that 191 were predicted as negative which were True negative but 4 predicted as negative which were True positive [11]. Likewise, 973 were predicted as positive which were True positive, but 9 were predicted as positive which were True negative.

Likewise, the confusion matrix for Multinomial NB(Naïve Bayes) shown in Fig.-7b Which represents that 146 were predicted as negative which were True negative but 17 were predicted as negative which were True positive. Likewise, 960 were predicted as positive which were True positive, but 54 were predicted as positive which were True negative. As a result, Multi-layer Perceptron performs better than



Multinomial NB as is evident from the confusion matrices.

Fig. 7a Multi-Layer Perceptron's Confusion matrix

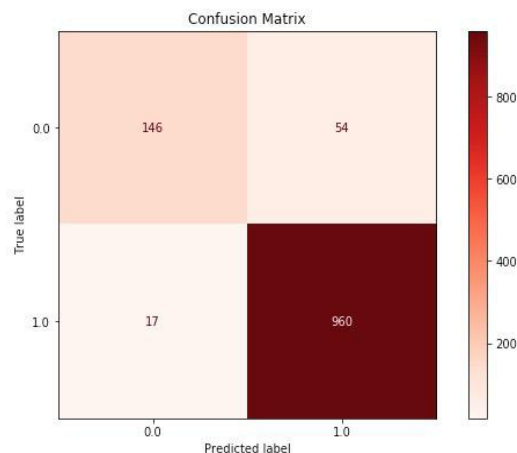


Fig. 7b Multinomial Naïve BayesConfusion matrix

We used a bar graph as shown in Fig. 8a to formalize our results obtained from confusion matrices to represent the variation in both the classification algorithms based on accuracy.

Also, performance is represented by a Receiver Operating Characteristics curve between True Positive Rate and False Positive Rate of both the algorithms shown in Fig. 8b. The ROC curve also shows that Multi-Layer Perceptron works slightly better than Multinomial Naïve Bayes.

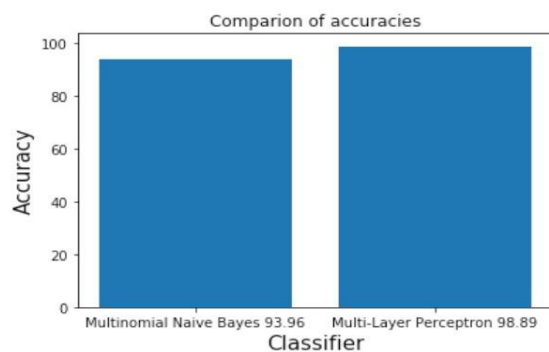


Fig. 8a Accuracy Comparison of algorithms shown by a Bar Graph

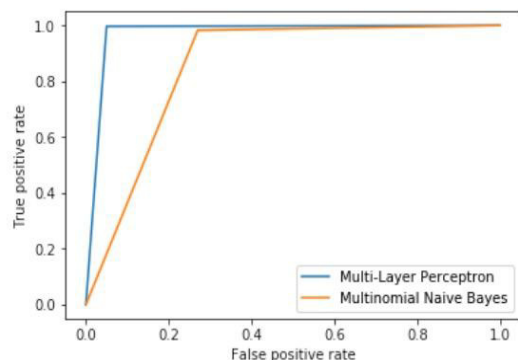


Fig. 8b TPR is greater for the Multi-Layer Perceptron classifier shown by ROC Curve.

After plotting the Bar Graph and ROC Curve, it is noticeable that Multi-Layer Perceptron has an accuracy of 98.89% which is greater than Multinomial NB (Naïve Bayes), which has an accuracy of 93.96% as shown in Fig. 8a and the former also provides better performance on the given dataset than the latter as shown by the Receiver Operating Characteristics curve

## V. CONCLUSIONS & FUTURE WORK

To conclude, from this research, Multi-Layer Perceptron achieves slightly better results than that of Multinomial Naïve Bayes on a dataset of automated product reviews with a higher number of reviews(1596 reviews in this case). Multi-Layer Perceptron touches an accuracy of about 98 percent is pretty efficient. Providing a higher amount of data, it will be easy to achieve high accuracies and vice versa for both the algorithms discussed above. Now, the authors also concluded that although Multi-Layer Perceptron touches greater accuracy, the variance in inaccuracies are not very remarkable as Multinomial NB(Naïve Bayes) also touches almost 93 percent of accuracy which shows that the performance of the two algorithms does not variate much for the provided dataset.

The future works include experimenting with different languages written in English, like in Indian E-commerce websites people usually feel more comfortable writing their reviews in their native language (i.e. Hindi). Therefore, by using a more advanced, accurate, and updated dataset for natural language processing methods and tools, this problem can be solved. Additionally, connotation understanding can also be integrated into the algorithm to generate a more accurate sentiment analysis. Some reviews are used sarcastically which look positive but have a negative review and vice versa.

## REFERENCES

- [1] Singla Z, Randhawa S and Jain S 2017 *Sentiment analysis of customer product reviews using machine learning*. International conference on intelligent computing and control (I2C2) 1-5.
- [2] Wahyudi M and Kristiyanti D A 2016 *Sentiment Analysis Of Smartphone Product Review Using Support Vector Machine Algorithm-Based Particle Swarm Optimization*. Journal of Theoretical & Applied Information Technology 91(1).
- [3] Singh G, Kumar B, Gaur L and Tyagi A 2019 *Comparison between multinomial and Bernoulli naïve Bayes for text classification*. International Conference on Automation, Computational and Technology Management (ICACTM) 593-596.
- [4] Fan X, Li X, Du F, Li X and Wei M 2016 *Apply word vectors for sentiment analysis of APP reviews*. 3rd International Conference on Systems and Informatics (ICSAI) 1062-1066.
- [5] Maas A, Daly R E, Pham P T, Huang D, Ng A Y and Potts C 2011 *Learning word vectors for sentiment analysis*. In Proceedings of the 49th annual meeting of the association for computational linguistics: Human language technologies 142-150.
- [6] Devi D N, Kumar C K and Prasad S 2016 *A feature based approach for sentiment analysis by using support vector machines*. IEEE 6th International Conference on Advanced Computing (IACC) 3-8.
- [7] Wang C and Wang F 2014 *Extracting sentiment words using pattern based Bootstrapping method*. Computer Engineering and Applications 50(1): 127-129.
- [8] Wiratama G P and Rusli A 2019 *Sentiment Analysis of Application User Feedback in Bahasa Indonesia Using Multinomial Naïve Bayes*.

- 5th International Conference on New Media Studies (CONMEDIA) 223-227.
- [9] Ravichandran K and Arulchelvan S 2017 *The model of multilayer perceptron analyzed the crime news awareness in India (quantitative analysis method)*. 4th International Conference on Advanced Computing and Communication Systems (ICACCS) 1-6.
  - [10] H. Bani-Salameh, S. M. Alkhatib, M. Abdalla et al., "Prediction of diabetes and hypertension using multi-layer perceptron neural networks," *International Journal of Modeling, Simulation, and Scientific Computing*, vol. 12, no. 2, 2021.
  - [11] Erevelles S, Fukawa N and Swayne L 2016 *Big Data consumer analytics and the transformation of marketing*. *Journal of business research* 69(2) 897-904.
  - [12] Sumian Peng, "Research on Data Preprocessing Process in the Web Log Mining" IEEE Conference. Information Science and Engineering (ICISE), 2009 1st International Conference, pp. 942 - 945, 26-28 Dec. 2009
  - [13] Sun Binli, "Research on data-preprocessing for construction of university information systems", IEEE Conference. Computer Application and System Modeling (ICCASM), 2010 International Conference Volume:1. 22-24 Oct. 2010
  - [14] Sudheer Reddy, K., "An effective data preprocessing method for Web Usage Mining", IEEE Conference Information Communication and Embedded Systems (ICICES), 2013 International Conference. 7 - 10 Feb. 2013.
  - [15] Wei Jianping, "Research on Data Preprocessing in Supermarket Customers Data Mining", IEEE Conference Information Engineering and Computer Science (ICIECS), 2010 2nd International Conference, pp. 1 - 4, 25-26 Dec. 2010.
  - [16] Qing Ang, "Explored research on data preprocessing and mining technology for clinical data applications", IEEE Conference. Information Management and Engineering (ICIME), 2010 The 2nd IEEE International Conference, pp 327 - 330. 16-18 April 2010.
  - [17] Sharon Christa, "Data Preprocessing Using Intelligent Agents", *Proceedings of the International Conference on Frontiers of Intelligent Computing: Theory and Applications (FICTA)*, pp 197-204.
  - [18] Amir R. Razavi, "A Data Pre-processing Method to Increase Efficiency and Accuracy in Data Mining", 10th Conference on Artificial Intelligence in Medicine, pp 434-443.
  - [19] Mikolov, T., Sutskever, I., Chen, K., Corrado, G.S., & Dean, J. (2013). *Distributed representations of words and phrases and their compositionality*. *Advances in Neural Information Processing Systems*, pp. 1-9
  - [20] Bui, D. A., & Zeng-Treitler, Q. (2014). "Learning regular expressions for clinical text classification," *J. Amer. Med. Inform. Assoc.*, vol. 21, no. 5, pp.850-857.